

Operating System (OS) Quiz

Answer

1. When will the multi-program system and the time-sharing system switch schedules?

A:

In a multiprogramming system, when a running task enters a waiting state, it switches to another task in memory to continue running. In addition to waiting for I/O, the time-sharing system's schedule may also be paused because the time quota is exhausted.

2. Suppose there is a computer system that does not provide system calls, but it does provide interrupt handling. How to simulate the mechanism of system calls on such a computer system?

A:

You can define an interrupt handling for system calls, and when users want to call system calls, they can use interrupt processing directly.

3. Please describe what are the usual statuses of the itinerary?

A:

Created: The state when the process was first created.

Ready: The itinerary is ready and ready to be executed.

Suspended: The journey is waiting for an event to occur, usually waiting for the I/O operation to complete.

Execution: The process is running, that is, the CPU is being used.

Terminated: The trip has ended, but some information remains in the system.

Displace/Ready: Procedures can be executed, but are currently overridden in secondary memory.

Displace/Suspend: The process is waiting for an event to occur and is overridden into secondary memory.

4. Please explain what is body switching?

A:

When CPU usage is switched from one process to another, the operating system must store information about the process in the program control section of that process and load the program control section of the other process into the system in order to restore the execution environment to the state it was in when the latter was interrupted. This action is called body switching.

5. In the multi-level queue scheduling method, what are the advantages if different queues have different time slices?

A:

In a multi-tiered queue scheduling method, the length of the time slice obtained for each queue can vary depending on the level of priority. For example, a queue with a higher priority will get a longer time slice, while a queue with a lower priority will get a shorter time slice. For example, an interactive foreground program may only need a short time slice, so there is no set rule that can regulate the length of time slices between queues, but must be determined according to the characteristics of process classification.

6. According to the instructions below, draw the final contents of the register when the command of memory addresses 200 to 206 is executed.

Instruction format :

opcode (4位元)	memory address (12位元)
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(A)List of opcodes :

opcode	Description	opcode	Description
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0000	Load data from memory into register A	0010	Subtract the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

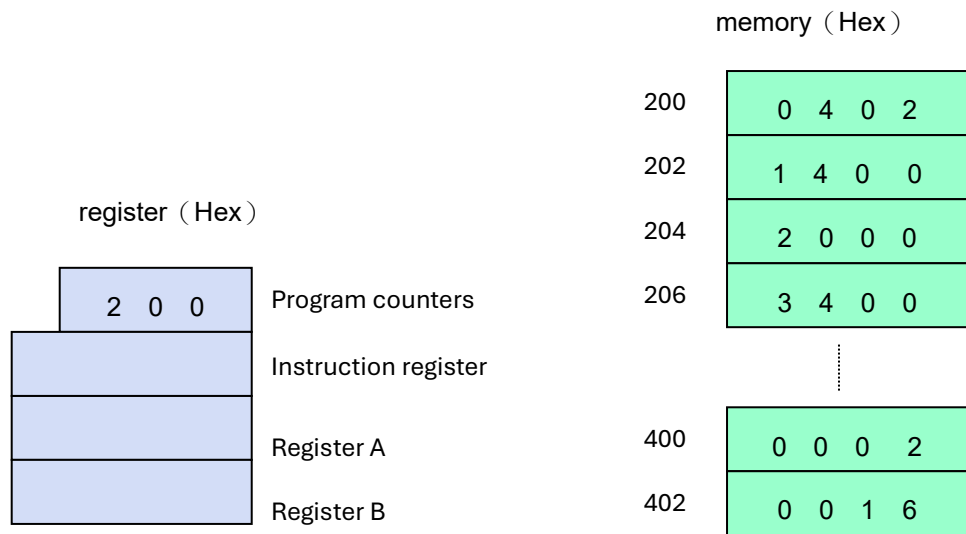
(B)List of opcodes :

opcode	Description	opcode	Description
0000	Load data from memory into register A	0010	Multiply the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

(C)List of opcodes :

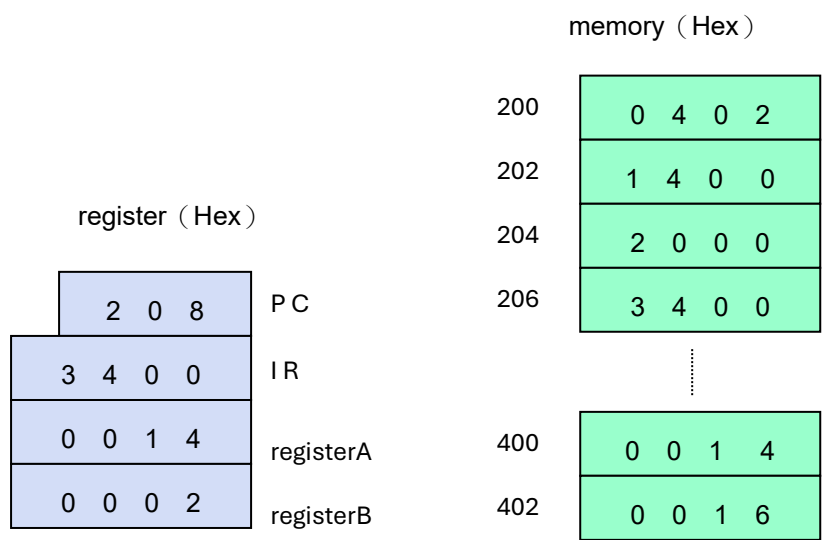
opcode	Description	opcode	Description
0000	Load data from memory into register A	0010	Divide the values of registers A and B and put them into register A
0001	Loads data from memory into register B	0011	Writes the value in register A to memory

Memory initial content :



A :

(A)



(B)

register (Hex)				
2 0 8				PC
3 4 0 0				IR
0 0 1 4				registerA
0 0 0 2				registerB

memory (Hex)			
200	0	4	0 2
202	1	4	0 0
204	2	0	0 0
206	3	4	0 0
.....			
400	0	0	1 4
402	0	0	1 6

(C)

register (Hex)				
2 0 8				PC
3 4 0 0				IR
0 0 1 4				registerA
0 0 0 2				registerB

memory (Hex)			
200	0	4	0 2
202	1	4	0 0
204	2	0	0 0
206	3	4	0 0
.....			
400	0	0	1 4
402	0	0	1 6